

Greedy Positive-Cone Reduction: from Three Baselines to a Standard Angular-Defect Greedy

Experimental summary

July 15, 2026

Abstract

We build a reduced *positive cone* $K = \text{span}_+\{w_1, \dots, w_R\}$ approximating a set of non-negative snapshots, and compare three greedy constructions — Cone-Projected Greedy (CPG), Angular-Defect Greedy (ADG), modified CPG (MCPG) — on a low-dimensional *lambda* dataset and a high-dimensional *physics* contact dataset. The note is organised as the *evolution* of the method: a fair baseline comparison exposes two weaknesses of the raw greedy cones (poor worst-case relative accuracy when snapshot magnitudes vary, and severe ill-conditioning from near-duplicate generators); we then add two ingredients to ADG — *per-snapshot normalisation* and an *angular redundancy* control $\varepsilon_\angle$ — and fold them into a **standard ADG**. At equal cone size the standard ADG is 3–5 $\times$ more accurate than CPG/MCPG on the wide-magnitude physics data, and on the well-scaled lambda data it turns an otherwise catastrophic condition number ($\sim 10^{10}$) into $\sim 10^1$.

1 Setup

Algorithms. All three methods grow a positive cone one generator (or, for ADG, one batch) at a time and project each snapshot x onto the current cone by non-negative least squares (NNLS), $\Pi_K(x) = \arg \min_{c \geq 0} \|Ac - x\|_2$ with A the generators.

- CPG adds the snapshot of largest *absolute* residual $\|x - \Pi_K(x)\|_2$.
- ADG adds the snapshot of largest *angle* $\theta_K(x) = \arcsin(\|x - \Pi_K(x)\|_2 / \|x\|_2)$ to the cone — a *scale-invariant* novelty measure, so its selection is intrinsically insensitive to snapshot magnitude.
- MCPG selects like CPG but stores a shifted, re-normalised generator $\nu = (x - \Pi_K(x)) / \|x - \Pi_K(x)\|_2$, carrying only the direction new to the cone.

Metric. We report the *per-snapshot relative residual* $\rho(x) = \|x - \Pi_K(x)\|_2 / \|x\|_2$, its worst case $\rho_{\max}$ and mean $\bar{\rho}$ over the dataset, evaluated on the *original*, *physical-scale* snapshots for every method, plus the basis condition number $\kappa_2 = \sigma_{\max} / \sigma_{\min}$.

Datasets. Table 1 lists the two datasets. The magnitude ratio $\max_x \|x\|_2 / \min_x \|x\|_2$ is the single quantity that predicts which method wins.

Table 1: Datasets. The magnitude ratio (over nonzero snapshots) governs the qualitative differences below.

Dataset	Snapshots $\times$ dim	$\min \ x\ _2$	$\max \ x\ _2$	Magnitude ratio
Lambda	50×57	0.283	0.319	1.13
Physics	96×7676	9.04×10^6	5.46×10^9	604

2 Baseline: the three methods as-is

Run at a common relative tolerance $\varepsilon = 10^{-2}$ (each method stops at its own cone size), the raw greedy cones already reveal the two problems that motivate everything else (Table 2).

Table 2: Baseline at $\varepsilon = 10^{-2}$ (whole dataset). Physics: ADG’s scale-invariant selection halves the worst-case error of CPG/MCPG. Lambda: MCPG is best and, crucially, CPG/ADG are numerically singular ($\kappa_2 \sim 10^{10}$).

Dataset	Method	R	$\rho_{\max}$	κ_2
Lambda	CPG	23	1.10×10^{-2}	5.43×10^{10}
	MCPG	15	8.90×10^{-3}	1.10×10^3
	ADG	23	1.10×10^{-2}	5.43×10^{10}
Physics	CPG	9	8.14×10^{-1}	2.10×10^2
	MCPG	9	8.06×10^{-1}	2.18×10^2
	ADG	11	3.53×10^{-1}	1.39×10^3

Two weaknesses. (i) **Accuracy on wide-magnitude data.** On physics, CPG/MCPG rank candidates by *absolute* residual, so they chase a few high-magnitude snapshots and leave the small-magnitude ones badly resolved ($\rho_{\max} \approx 0.8$); ADG’s angle criterion already does better (0.35), but not enough. (ii) **Conditioning.** On lambda the snapshots are nearly collinear, so raw-snapshot cones (CPG, ADG) accumulate near-parallel generators and become singular ($\kappa_2 \approx 5 \times 10^{10}$). The next two ideas target these directly, on ADG.

3 Idea 1 — per-snapshot normalisation

ADG’s *selection* is scale-invariant, but its *stopping* rule still compares the absolute residual to a single global scale $\varepsilon \max_x \|x\|$. On wide-magnitude data this lets a small snapshot leave the “unresolved” set while still 50%+ wrong in relative terms. Measuring the error relative to each snapshot’s own norm (dividing x by $\|x\|$ before the stopping check, while still storing physical-scale generators) makes the criterion a genuine per-snapshot relative one. It is applied to ADG *only*; CPG/MCPG have no such option and stay on the raw snapshots.

Figure 1 shows the effect on physics: the raw methods leave a broad plateau of small-magnitude snapshots stuck near 10^{-2} relative error, while normalised ADG drives them to machine precision. At $\varepsilon = 10^{-2}$ normalised ADG reaches $\rho_{\max} = 0.143$ (mean 3.7×10^{-3}) versus 0.35 for raw ADG and 0.81 for CPG/MCPG. On lambda (uniform norms) normalisation is inert.

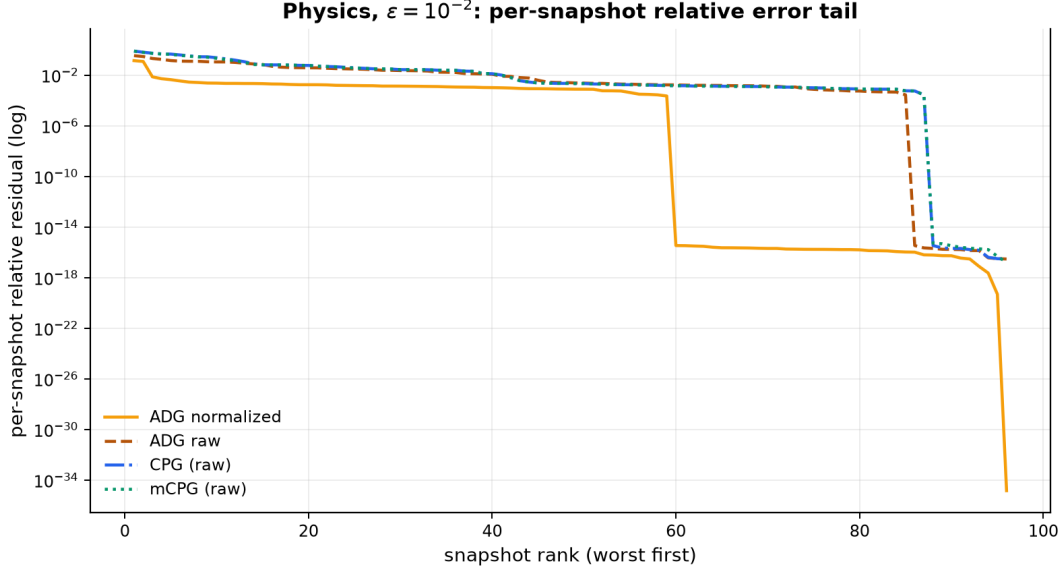


Figure 1: Physics, $\varepsilon = 10^{-2}$: per-snapshot relative residual sorted worst-first (log scale). Normalisation (applied to ADG only) resolves the small-magnitude snapshots that a global-scale criterion neglects, which is what collapses $\rho_{\max}$.

4 Idea 2 — angular redundancy control

Normalisation improves accuracy but not conditioning or cone size. On nearly collinear data ADG keeps adding generators that point almost the same way, inflating R and κ_2 . The redundancy control adds a second tolerance $\varepsilon_{\angle}$: a candidate is admitted only if its pairwise angle to *every* existing generator exceeds $\arcsin \varepsilon_{\angle}$, and enrichment stops once every remaining unresolved snapshot is redundant. No two generators can then lie within $\arcsin \varepsilon_{\angle}$ of each other, so R and κ_2 shrink at a controlled accuracy cost (Table 3). It can only *keep or diminish* R ; $\varepsilon_{\angle} = 0$ recovers plain ADG.

Table 3: Angular redundancy on ADG (lambda, $\varepsilon = 10^{-2}$, normalisation on). The knob trades a little worst-case accuracy for a far smaller, dramatically better-conditioned cone.

ADG variant	R	$\rho_{\max}$	κ_2
plain ($\varepsilon_{\angle} = 0$)	23	1.10×10^{-2}	5.43×10^{10}
$\varepsilon_{\angle} = 0.05$	7	3.95×10^{-2}	1.13×10^2
$\varepsilon_{\angle} = 0.1$	2	7.59×10^{-2}	1.48×10^1

5 Idea 3 — angular batching (a cost lever)

A third, optional knob $\varepsilon_{\angle}^{\text{batch}}$ admits, per round, a *pairwise-separated batch* of the near-maximal-angle candidates instead of one. It leaves R and accuracy essentially unchanged but cuts the number of greedy rounds several-fold (e.g. physics $9 \rightarrow 5$, a smooth synthetic set $41 \rightarrow 4$), which matters when the per-round NNLS projection dominates cost. It is orthogonal to the two ideas above and off by default.

6 The standard ADG and a fair comparison

We define the **standard ADG** as plain ADG with per-snapshot normalisation on and the redundancy control $\varepsilon_{\angle}$ as a single knob (default 0.05). To compare it fairly against CPG/MCPG we let the standard ADG choose the cone size R (via $\varepsilon_{\angle}$), then fit CPG and MCPG to the *same* R and compare — a matched-cone-size head-to-head (Figures 2, 3; Table 4).

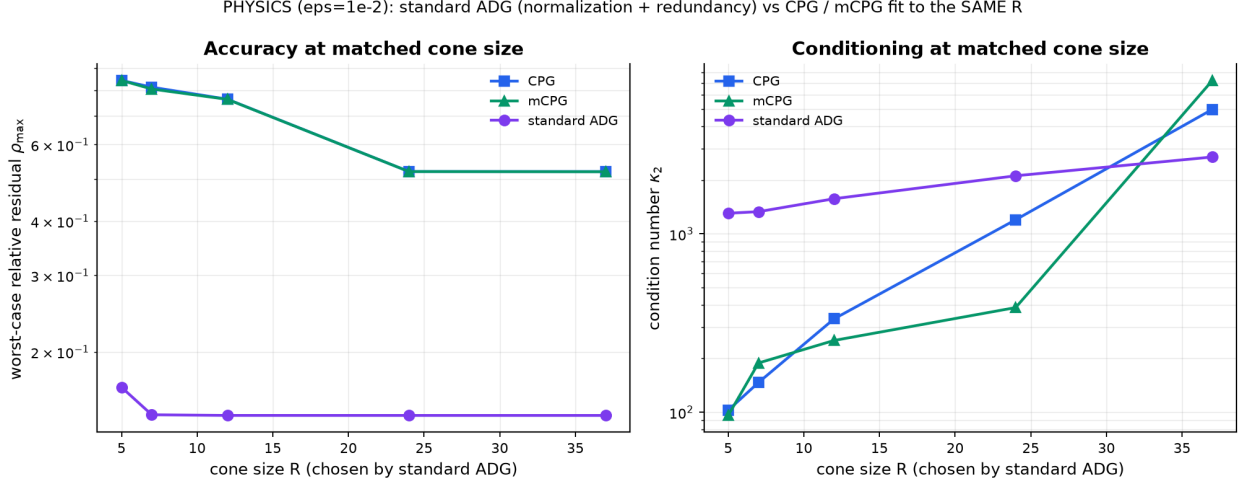


Figure 2: Physics: standard ADG versus CPG/MCPG fit to the *same* cone size R (chosen by ADG’s redundancy sweep). At every R the standard ADG’s worst-case relative error is 3–5 $\times$ lower; its condition number grows slowly while CPG/MCPG blow up past $R \approx 30$.

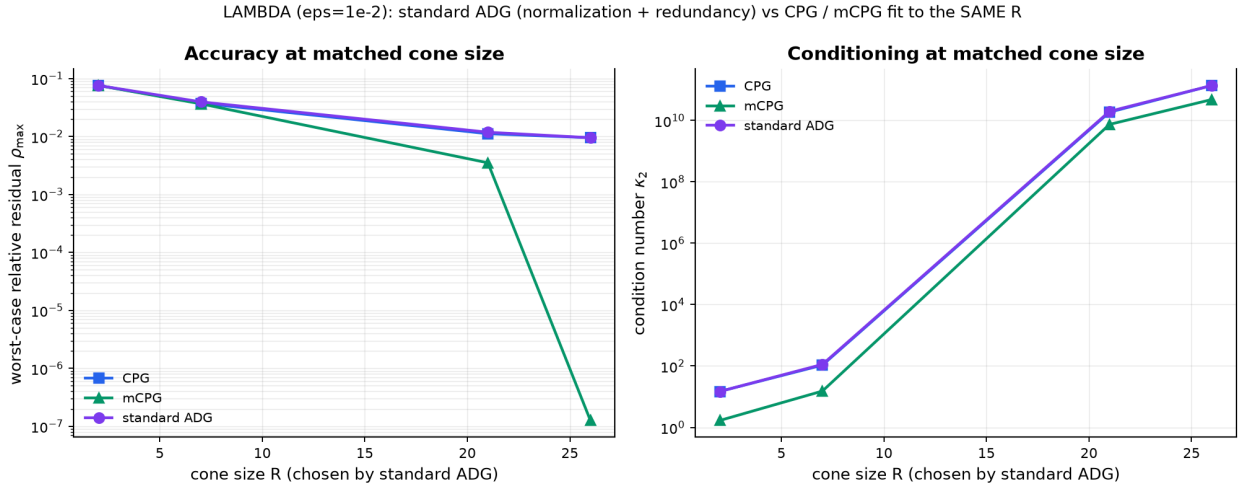


Figure 3: Lambda: same matched- R comparison. On well-scaled data the three are close on accuracy (MCPG edges ahead and stays best-conditioned); the standard ADG’s value here is a compact, well-conditioned cone rather than the lowest error.

Reading the comparison. On **physics** the standard ADG dominates: at any cone size its worst-case relative error is ≈ 0.14 while CPG/MCPG sit at 0.52–0.84, because only ADG (through

Table 4: Matched cone size: worst-case relative residual $\rho_{\max}$ for the standard ADG (normalisation + redundancy) and for CPG/MCPG fit to the same R ($\varepsilon = 10^{-2}$, whole dataset).

Dataset	R	ADG $\rho_{\max}$	CPG $\rho_{\max}$	MCPG $\rho_{\max}$
Physics	5	1.66×10^{-1}	8.43×10^{-1}	8.43×10^{-1}
	7	1.43×10^{-1}	8.14×10^{-1}	8.06×10^{-1}
	12	1.43×10^{-1}	7.64×10^{-1}	7.63×10^{-1}
	24	1.43×10^{-1}	5.20×10^{-1}	5.20×10^{-1}
Lambda	2	7.59×10^{-2}	7.59×10^{-2}	7.59×10^{-2}
	7	3.95×10^{-2}	3.75×10^{-2}	3.66×10^{-2}
	21	1.18×10^{-2}	1.12×10^{-2}	3.50×10^{-3}
	26	9.52×10^{-3}	9.52×10^{-3}	1.00×10^{-6}

normalisation) attends to the small-magnitude snapshots. On **lambda** the three methods are close on accuracy at matched R ; MCPG is the best accuracy-per-component method, while the standard ADG’s contribution is the redundancy knob that yields a compact, well-conditioned cone (Table 3).

7 Conclusions

- **Baseline.** Raw greedy cones have two failure modes: poor worst-case relative accuracy when snapshot magnitudes vary (CPG/MCPG on physics), and near-singular bases from redundant generators (CPG/ADG on lambda).
- **Normalisation** fixes the first for ADG: on physics it cuts the worst-case relative error from 0.35 (raw ADG) / 0.81 (CPG/MCPG) to 0.14; it is inert on uniform-magnitude data.
- **Angular redundancy** fixes the second: it shrinks R and turns the lambda cone’s $\kappa_2 \approx 5 \times 10^{10}$ into $\sim 10^2$ ($\sim 10^1$ at $R = 2$), for a modest accuracy cost, and reduces R for free on physics.
- **Standard ADG** (normalisation + redundancy) is best on wide-magnitude data at every matched cone size (3–5 $\times$ lower $\rho_{\max}$ than CPG/MCPG); on well-scaled data MCPG leads on accuracy while the standard ADG offers a compact, well-conditioned alternative. *Guidance:* use the standard ADG for data with a wide spread of snapshot magnitudes or when a small, well-conditioned cone matters; use MCPG for well-scaled data where error-per-component is paramount; add the batching knob when the round count is the bottleneck.

Reproducibility. Baseline / standard-ADG comparison: `results/method_comparison_redundancy/` (`new_adg_vs_cpg_mcpg.csv`, `matched_R_metrics.csv`, `matched_R_{lambda,physics}.png`). Normalisation study: `results/physics/adg_normalization/` and `results/lambda/adg_normalization/`. Per-run standard-ADG figure suites: `results/lambda/adg_standard/` and `results/physics/adg_standard/`. All numbers are taken verbatim from the `*.csv` and report files there.